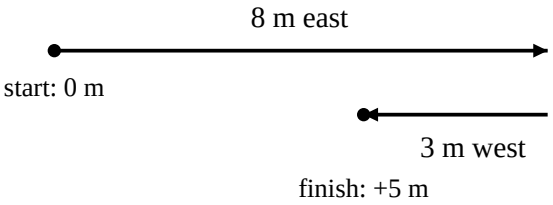
Describing Motion

1. Distance and displacement

Distance: the whole path. **Displacement:** finish - start.



Distance = 8 + 3 = **11 m**
Displacement = 5 - 0 = **5 m east**

2. Speed and velocity

$$\text{average speed} = \frac{\text{total distance}}{\text{total elapsed time}}$$

Example: 150 m in 100 s
150 ÷ 100 = **1.5 m/s**

Distance = speed × time **Time = distance ÷ speed**

2 m/s east

2 m/s west

Same speed. Different velocities because direction is different.

3. Acceleration: a change in velocity

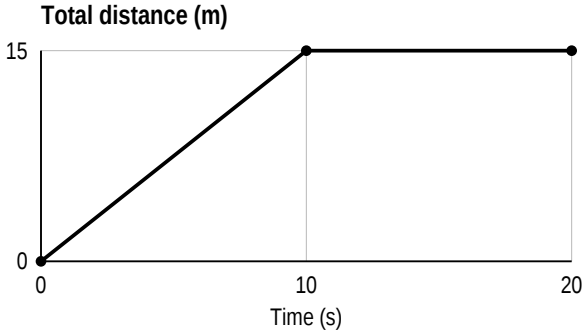
$$a = \frac{\text{final velocity - initial velocity}}{\text{elapsed time}}$$

Elapsed time = final time - initial time.
From 2 s to 6 s: 6 - 2 = 4 s.

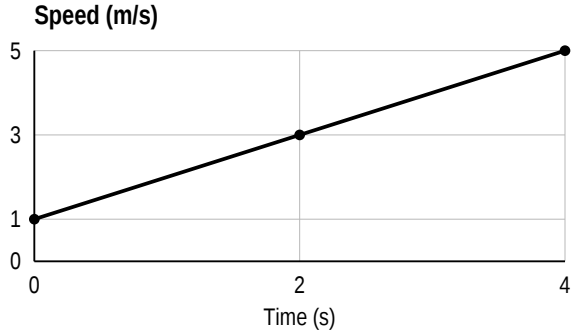
Example: a cyclist goes east from 3 m/s to 9 m/s in 4 s.
1. Find the change: 9 - 3 = 6 m/s
2. Divide: 6 ÷ 4 = 1.5 m/s² east

v_i = initial velocity; v_f = final velocity. $\mathbf{v_f = v_i + a \times}$ **elapsed time.**
Acceleration opposite velocity slows an object. A minus sign shows direction, not always slowing.

4. Read the axes before reading the graph



Slope = speed. Flat means stopped.
First 10 s: 15 ÷ 10 = **1.5 m/s**.

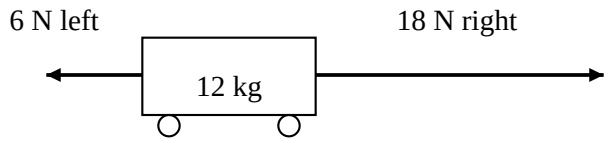


For straight motion in the positive direction:
(5 - 1) ÷ 4 = **1 m/s²**.

A speed graph does not show direction. Constant speed can still include turning.

Forces and Momentum

1. Combine forces, then find acceleration

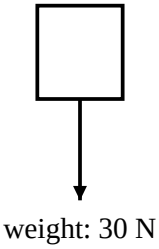


F_{net} = ma; a = F_{net} ÷ m
m = F_{net} ÷ a
1. F_{net} = 18 - 6 = **12 N right**.
2. a = 12 ÷ 12 = **1 m/s² right**.

Zero net force: rest or constant velocity. **Inertia:** resistance to a change in motion. More mass means more inertia. Use net force, not just one push or pull.

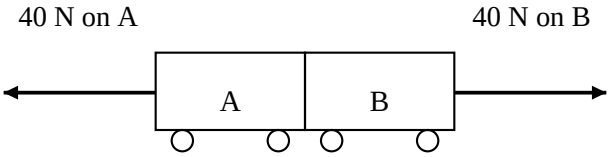
2. Gravity and friction

Weight = mass × g Earth: g ≈ 10 m/s²
A 3 kg bag weighs 3 × 10 = **30 N**.
Mass is measured in **kg**; weight in **N**.



Friction opposes sliding. Static friction adjusts to the push, up to its limit.
A 7 N push on a resting crate is balanced by **7 N** of friction if its limit is higher.

3. Action and reaction act on different objects

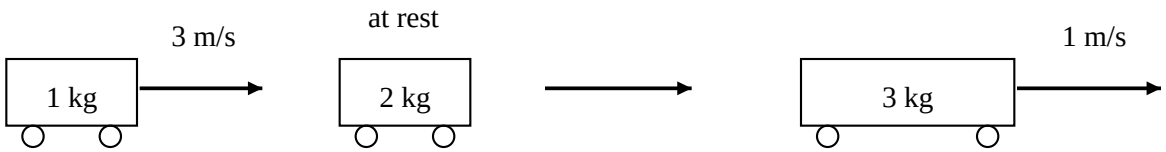


During a push; ignore friction.
Equal forces act on different carts.
With equal net forces: lighter means more acceleration.

Turning at steady speed: velocity changes direction. Net force points toward the center; tire-road static friction can supply this inward force.

4. Momentum and carts that stick together

p = mass × velocity Momentum unit: kg·m/s



When external forces have negligible effect during the brief collision:

1. Add momentum: (1 × 3) + (2 × 0) = **3 kg·m/s**.
2. Add the masses: 1 + 2 = **3 kg**.
3. Shared velocity = total momentum ÷ total mass = 3 ÷ 3 = **1 m/s right**.

Every calculation: formula → substitute → calculate → unit and direction.